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Research Article

Scale-Dependent Techno-Economic Analysis of CO_2 Capture and Electroreduction to Ethylene

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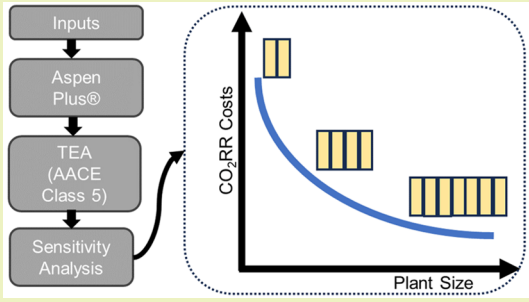
ABSTRACT: The decarbonization of the chemical industry is essential to mitigate carbon dioxide (CO_2) emissions. Ethylene (C_2H_4) is the highest production petrochemical globally. When powered by renewable electricity, the electrochemical conversion of CO_2 to C_2H_4 offers a promising route to low carbon C_2H_4 production. We perform a detailed techno-economic assessment (TEA) of the CO_2 reduction reaction (CO_2RR) process, converting CO_2 from an industrial point source to polymer-grade C_2H_4. We pair the CO_2 electrolyzer with industrially mature upstream and downstream separation technologies in an Aspen Plus model. This comprehensive approach enables us to assess the valorization of both gas and liquid byproduct streams at commercial specification and assess the viability of these processes as a function of scale. We demonstrate that a minimum plant size of ∼3,000 tonne C_2H_4/year is needed to achieve economies of scale among the upstream and downstream processes. This minimum plant size is ∼200-fold smaller than that of conventional C_2H_4 plants, coincides with that of typical utility-scale solar installations (∼25 MW), and could enable a more distributed model of chemical production going forward. We further highlight technical and economic enablers that would increase the profitability of the CO_2RR to C_2H_4 technology.



KEYWORDS: *decarbonization, techno-economic analysis, CO_2 electroreduction, CO_2 electrolysis, distributed production, renewable electricity, Aspen Plus*

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■ INTRODUCTION

Ethylene (C_2H_4) is a precursor for many materials such as polyethylene, ethylene glycol, poly(vinyl chloride), and polyethylene terephthalate.[1,2] C_2H_4-derived products are used extensively throughout the packaging, construction, and automotive industries.[3] Since 2014, 120 billion USD has been allocated to building new C_2H_4 plants, and in 2020, the global production capacity was 200 million tonnes with a 6% year-over-year growth rate.[4,5] C_2H_4 is produced via the steam cracking of naphtha and natural gas. However, this conventional production method is carbon intensive (2−4 tonne CO_2/tonne C_2H_4)[1,6] and contributes substantially to anthropogenic carbon dioxide (CO_2) emissions.[7,8]

Alternative C_2H_4 production strategies have been proposed, such as C_2H_4 production via the dehydration of biomass-derived ethanol (C_2H_5OH)[9,10] and the replacement of polyethylene with bioplastics based on poly(lactic acid)s. The dehydration of biomass-derived C_2H_5OH has received only limited investment to date, as insufficient biomass availability may prevent this route from satisfying market demand.[4,11] Plastics made from poly(lactic acid)s are also subject to biomass availability concerns and differ in thermal/mechanical/usage specifications when compared to polyethylene, making direct material substitution a challenge.[12]

When powered with low carbon renewable electricity, the electrochemical CO_2 reduction reaction (CO_2RR) offers a means to manufacture C_2H_4 with much lower carbon emissions.[13,14] CO_2 electroreduction can produce C_2H_4 that is identical with that derived from fossil fuel steam cracking and compatible with the existing petrochemical downstream processes. Recent CO_2RR demonstrations have achieved high selectivity toward C_2H_4, extended stability, increased electrolyzer size, and high current density operation.[15−19] Industrial implementation of the CO_2RR will require integration with upstream and downstream separation technologies, and each

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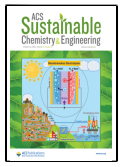

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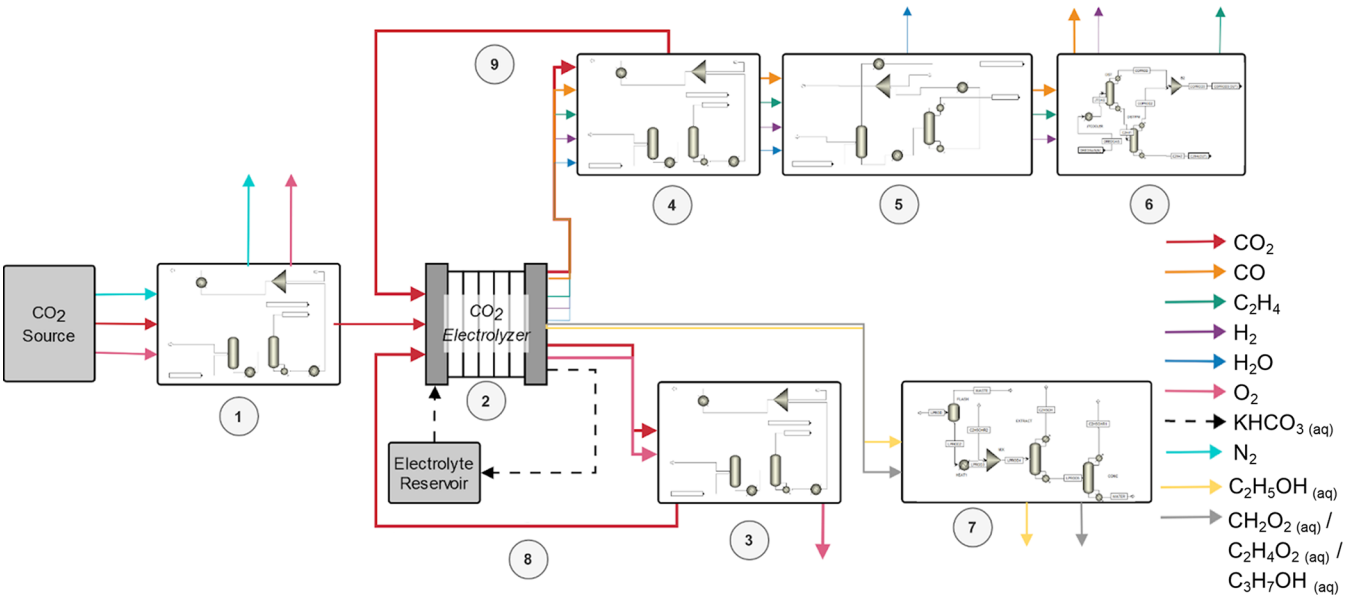

Figure 1. Complete process flow diagram from the CO_2 point source to polymer grade C_2H_4 with syngas (H_2/CO), O_2, and C_2H_5OH coproduction via CO_2RR. The process is comprised of 9 stages: (1) upstream CO_2 capture (MEA absorption), (2) CO_2 electrolysis (AEM or BPM zero-gap electrolyzer), (3) anodic CO_2 removal (MEA absorption), (4) cathodic CO_2 removal (MEA absorption), (5) cathodic H_2O removal (TEG absorption), (6) cathodic C_2H_4 purification (cryogenic distillation), (7) cathodic C_2H_5OH purification (double distillation), (8) anodic CO_2 recycle, and (9) cathodic CO_2 recycle.

stage in this process will present economic and environmental considerations.

Prior techno-economic assessments (TEAs)[20−22] have provided order-of-magnitude estimates[23,24] to identify the most important electrolysis performance metrics and CO_2RR products—all at a hypothetical industrial scale. Our prior work modeled C_2H_4 separation downstream of a 100 m^2 CO_2 electrolyzer in Aspen Plus to quantify the impact of key electrolysis parameters on the separation energy penalty.[25] However, the costs of chemical processes are typically scale-dependent, and these dependencies are distinct for each unit process. Although most of these TEAs have shown a path to improve economics, these analyses have focused on the electrolyzer with only high-level assumptions made regarding the upstream and downstream separation units—operations that significantly influence costs and system performance in practice.[26,27] The mixtures produced from the CO_2RR system also differ in physicochemical properties from conventional (petro)chemical streams, and thus the underlying assumptions for the costs of unit operations may be inaccurate and introduce additional sources of uncertainty. These uncertainties are compounded in complex integrated systems with multiple subsystems that are the norm in chemical production.

This work presents a comprehensive scale-dependent assessment of integrated C_2H_4 production via CO_2RR, using equipment sizing and equations of state with the aim to inform, derisk, and inspire industrial implementation. We employ an integrated CO_2RR electrolyzer with mature downstream gas separation processes using the Aspen Plus model we developed previously.[25] We expand this model by adding high technology readiness level (TRL) separation technologies for upstream CO_2 capture as well as downstream liquid purification to achieve a fully integrated process. The boundary of the analysis begins at an industrial CO_2 point source and ends at polymer-grade C_2H_4 (99.9 wt % at −104 °C) with liquid and gas coproduct generation. We then perform a TEA of the integrated CO_2RR process using detailed cost estimations and a conservative factorial method that accounts for sizing and factors typical of chemical plant designs.[23,24,28] We analyze the economic potential of the CO_2RR to C_2H_4 as a function of electrolyzer configuration and scale, identifying the most costly stages in the overall process. Finally, we perform a sensitivity analysis and identify opportunities for improvements in profitability.

■ MATERIALS AND METHODS

Process Overview. The complete CO_2 point source to the polymer grade C_2H_4 process was modeled in Aspen Plus v10 (Aspen Tech Inc., Bedford, MA) in nine stages (Figure 1): the capture and purification of CO_2 from an upstream industrial CO_2 point source; the conversion of CO_2 to C_2H_4 and coproducts in the electrolyzer; the separation of the different cathodic and anodic products to commercial grade chemicals (three stages for cathode gases, one for anode gases, and one for cathode liquids); and the recycling of unreacted cathodic and anodic CO_2 (two stages). For the upstream and downstream separation units, we qualitatively screened available technologies based on TRL ranking[29,30] and selected ones with strong operational history and validated modeling data (Supporting Information Note 1). Each separation stage was assigned to an equation of state according to the literature (Supporting Information Note 2).[31−38] C_2H_4 was the main product, but syngas (a mixture of carbon monoxide (CO) and hydrogen (H_2)), oxygen (O_2), and C_2H_5OH were valorized as coproducts.

Upstream CO_2 Capture. The feedstock for the process was flue gas, containing CO_2 at 6.4 wt %, a typical concentration in flue gas streams from aluminum producers, natural gas power plants, and petroleum refineries.[39,40] The flue gas also contained 80.0 wt % nitrogen (N_2) and 13.7 wt % oxygen (O_2).[41] CO_2 in this stream was captured by a monoethanolamine (MEA) absorption-based unit[31−33] which concentrated and humidified the CO_2 (fully humidified CO_2 at 28 °C, 98.2 wt % CO_2, and 1.8 wt % water (H_2O)) for the CO_2 electrolyzer.

CO_2 Electrolyzer. To represent the CO_2RR technology, two electrolyzer configurations were defined (Supporting Information Note 3). The first configuration was based on lab-scale demon-

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strations with an anion exchange membrane (AEM) with some assumed losses in performance (e.g., voltage and selectivity) as the electrolyzer was scaled-up. This electrolyzer configuration suffered CO_2 crossover from the cathode to anode such that the anodic tail gas was a CO_2−O_2 mixture.[42] The anolyte was a potassium bicarbonate ($KHCO_3$) solution which, although recirculated through the electrolyzer, requires the occasional top up due to salt precipitation in the electrolyzer.[42] In the second configuration, a bipolar membrane (BPM) was employed to eliminate CO_2 crossover and achieve higher CO_2 utilization.[43,44] This BPM configuration was also based on current lab-scale performance but assumed no losses as the electrolyzer size was increased. It was assumed that the BPM configuration operated without cations in the anolyte and instead operated on a deionized (DI) water anolyte. Electrolyzer performance (e.g., CO_2 utilization, selectivity, voltage, and current density) may change as a function of scale,[19,45] but as a simplification, the electrolyzer performance remained constant for all production scales in this work. In both electrolyzer configurations, the evolution of O_2 occurred at the anode.

Humidified CO_2 was reacted over an electrocatalyst-coated cathode to produce gas products, such as CO and C_2H_4, as well as liquid products, such as C_2H_5OH, formic acid (CH_2O_2), acetic acid ($C_2H_4O_2$), and 1-propanol (C_3H_7OH). The parasitic coproduct H_2 and unreacted CO_2 were mixed with the cathode gas products. It was assumed that all liquid products could be collected from the cathode. There were three main output streams from the electrolyzer (Figure 1): a gaseous cathodic stream, a liquid cathodic stream, and a gaseous anodic stream.

Cathodic Gaseous Downstream Separation. The gaseous cathodic output stream was composed of a gaseous mixture of C_2H_4, H_2O, CO, H_2, and unreacted CO_2. The primary means of purifying C_2H_4 industrially is cryogenic distillation.[46] To avoid freezing during this process, CO_2 and H_2O must be removed beforehand.[47] Consequently, the cathode gas stream was purified in three sequential stages: CO_2 removal, H_2O removal, and C_2H_4 purification.

The removal of CO_2 from the cathodic tail gas stream is discussed in detail in the next section. The mixture of C_2H_4, H_2O, CO, and H_2 was then dehydrated using a triethylene glycol (TEG) absorption-based unit.[34] Finally, a double cryogenic distillation[35] yielded two separate streams: a cryogenic liquid polymer grade C_2H_4 (99.9 wt % at −104 °C) ready for storage and shipment[48] and a syngas coproduct stream.

Cathodic CO_2 Removal. In the first stage of cathodic gaseous downstream separation, unreacted CO_2 was separated from the cathodic gas products using an MEA carbon capture unit.[31−33] We simulated a range of C_2H_4 production rates and found that, at higher production rates (>1,000 tonne C_2H_4/year), the model predicted significant C_2H_4 absorption in the amine solution of the cathodic gas separation unit (Figure S5). For example, using a MEA capture solution on the BPM tail gas at 3,150 tonne C_2H_4/year, 40% of the C_2H_4 was absorbed in the amine solution and recycled back to the electrolyzer with the recovered CO_2. This absorption not only diluted the CO_2 recycle feed but also lowered the net C_2H_4 production rate. We modeled diglycolamine, another amine-based CO_2 capture solution, but the problem persisted, with even more C_2H_4 lost to the downstream CO_2 recovery solvent and onset at a lower C_2H_4 production rate (Figure S5). These findings suggested that the highest TRL CO_2 recovery technology (amine absorption) may not be suitable for the downstream separation of the CO_2RR at large scales. The model predicted such behavior based on the solubility data currently available within Aspen Plus's database, so experimental validation of the vapor liquid equilibrium data for this particular mixture would further strengthen the model.

We simulated a Benfield solution (e.g., potassium carbonate) as an alternative method for the recovery of CO_2 from this stream. At low absorption temperatures, less than 0.56% of the incoming C_2H_4 was lost to the CO_2 recovery solvent at all scales simulated, suggesting that this may be a suitable alternative for this separation process (Figure S5). Another option could be to change the system architecture to a tandem system wherein CO_2 is first reduced to CO and then reduced to C_2H_4. This approach limited the complexity of the cathodic gaseous mixture, and our simulations suggested no loss of CO to the downstream CO_2 recovery solutions (Figure S6). In addition, the development of separation technologies that do not rely on absorption mechanisms could offer other means of separating CO_2 from the cathodic gas product stream.

We modified the simulation scheme to virtually circumvent C_2H_4 absorption by the MEA solution at high throughputs (e.g., >1,000 tonne C_2H_4/year) and estimate the separation costs. CO_2 was diluted in a mixture of N_2/O_2 (5.8:1 N_2/O_2, a mass ratio commonly found in flue gas[41]), to be the same CO_2 mass fraction as the cathodic tail gas. Running the MEA carbon capture unit on this feedstock, we were able to collect the leftover CO_2 and assess the process costs, in terms of consumables and energy. All other constituents of the cathodic tail gas stream were assumed to pass through the MEA solution without loss. Employing our modified separation method at high production rates, we limited C_2H_4 loss to less than 5% at all scales—note that these losses do not originate from the CO_2 removal stage (Figure S7). Since this modified method was employed at scales >1,000 tonne C_2H_4/year, the economic estimation was subjected to associated uncertainties in this range. In the absence of experimental data, we postulate that this modification yielded an acceptable estimate of practical CO_2 capture requirements.

Cathodic C_2H_5OH Purification. The cathodic liquid stream consisted of an aqueous mixture of C_2H_5OH, CH_2O_2, $C_2H_4O_2$, and C_3H_7OH. C_2H_5OH was considered as a coproduct due to its modest CO_2RR selectivity and high market value. C_2H_5OH was extracted at 95 wt % via double distillation[36−38] from the azeotropic mixture of H_2O, CH_2O_2, $C_2H_4O_2$, and C_3H_7OH. The remaining liquid species were considered as waste.

Anodic Gas Purification. The gaseous anodic stream was composed of a humidified O_2 and a humidified CO_2−O_2 mixture in the BPM and AEM configurations, respectively. This stream could have been vented, at the expense of increased upstream capture energy and more CO_2 emissions for the AEM configuration (thereby lessening the environmental benefit of the technology, compared to fossil incumbent).[19] However, we quantified the financial burden to recover the CO_2 using an MEA carbon capture unit and considered O_2 as a coproduct.[31−33] Instead of mixing the anodic stream with the cathodic stream, we modeled a separate MEA carbon capture unit to avoid operational risks generated from a combustible oxyfuel mixture (i.e., H_2 from the cathode and O_2 from the anode).[49] The separation of the anodic stream via the MEA carbon capture unit yielded two streams: CO_2 (∼98 wt %) and O_2 (∼98 wt %). In the case of the BPM configuration, no anodic downstream CO_2 separation was required, but a simple flash step was required to remove water content from the O_2 stream.

Anodic and Cathodic CO_2 Recycle. The pure CO_2 recovered from the cathode and anode gas streams was extracted at 20 °C. These streams were heated to 28 °C prior to being recycled back to the electrolyzer input to match the inlet temperature specification of the electrolyzer.

Cost Estimation. From the process simulation, we exported the overall mass balance, the parameters and equipment models, the optimized and integrated energy balance (from the Aspen Plus automated pinch analysis), and the equipment sizing (from the Aspen Plus Icarus module). These exports were the key inputs for the techno-economic inventory phase (Supporting Information Note 4) and the associated determination of capital expenditures (CAPEX), operating expenditures (OPEX), revenues, net present value (NPV), and total annualized cost (TAC).[23,24,50] TAC is the annualized process cost, accounting for depreciation, whereas NPV represents the capital loss (negative value) or gain (positive value) expected after building and operating the plant for the specified lifetime.

We considered C_2H_4 as the main product, with syngas (CO/H_2 mixture), O_2, and C_2H_5OH as coproducts. The production costs can be assigned exclusively to C_2H_4 as the main product (C_2H_4 only), distributed according to the mass fraction of C_2H_4 among all coproducts (mass allocation), or distributed according to the revenue fraction of C_2H_4 relative to overall revenue (market allocation).

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However, by definition, the NPV was normalized only by the C_2H_4 production rate, although it did take into account the revenue generated from the other coproducts.

The cost estimation method was based on the factorial method designed by IFP/Technip[24] with the value expressed in USD (2019). The methodology used is in line with TEA guidelines.[50] Based on the data quality, the cost estimate was American Association of Cost Engineering Class 4 for the upstream and downstream separation equipment and Class 5 for the electrolyzer since it was based on experiments rather than kinetic models.[28] The overall accuracy of this study was then estimated to be between −30% and +50%.

The cost estimation was subject to external parameters, such as the prices of utilities, consumables, and (co)product market prices (Table 1). With a CO_2 supply cost of 0 USD/tonne, as is typical for a point source, there was no cost (e.g., for CO_2 transport to the upstream CO_2 capture unit) or value (e.g., from carbon pricing/emissions trading) assigned to the CO_2 in the base case scenario. The materials of construction required for CO_2 electrolyzers are, at the present, not well-established, and neither are the associated CAPEX. Therefore, it was assumed that the normalized electrolyzer cost remained constant at all scales, although we would expect some benefit from manufacturing economies of scale as seen in water electrolyzers.[51]

Table 1. Key Economic Assumptions[a]

Parameter	Value	Unit
CO_2 supply	0	USD/tonne
Electricity	12	USD/MWh
Lang factor	4.05	-
Declining balance depreciation factor	2	%
Tax rate	10	%
Interest rate	10	%
Plant lifetime	20	years
Fixed costs	9	% of CAPEX
C_2H_4 market price	967	USD/tonne
C_2H_5OH market price	1250	USD/tonne
Syngas market price	15.5	USD/GJ
O_2 market price	500	USD/tonne
Electrolyzer cost	1840	USD/m^2

[a]A complete list of economic assumptions and associated references is provided in Supporting Information Note 4.

RESULTS AND DISCUSSION

Cost Dependency on CO_2 Electrolyzer Configuration. We began the analysis by comparing the AEM and BPM CO_2RR configurations at 50,000 tonne C_2H_4/year. At this scale, the upstream CO_2 capture rate for the AEM configuration (995,000 tonne CO_2/year) is similar to the largest postcombustion CO_2 capture units demonstrated to date and 2% of the global carbon capture capacity—the vast majority of which is from point sources.[52,53] With an NPV of −213 USD/tonne C_2H_4, the AEM configuration was not profitable (Figure 2a). Changing to the BPM configuration enabled profitability with an NPV of 754 USD/tonne C_2H_4. Petrochemical C_2H_4 profits can vary with time, location, and cracker feedstock, but in 2019, the profitability of North American and Western European C_2H_4 producers was ∼200 and ∼500 USD/tonne C_2H_4, respectively.[54,55]

The TAC decreased substantially with the shift from the AEM to the BPM irrespective of the allocation method (Figure 2b). When all costs are attributed to C_2H_4, the TAC fell from 19,200 to 4,070 USD/tonne C_2H_4. The market and mass allocations lowered the TAC greatly (mass allocation more so than market allocation) by distributing the costs among the different coproducts: the BPM configuration had a market- and mass-allocated TAC of 1,040 and 598 USD/tonne C_2H_4, respectively. In 2019, petrochemical C_2H_4 was produced at a cost of 450−600 USD/tonne C_2H_4.[55]

C_2H_4 accounted for a greater product fraction in the BPM configuration when compared to the AEM configuration (Figure 2c). C_2H_4 accounted for 26% of the product revenue and 15% of the product mass in the BPM configuration. By comparison, C_2H_4 accounted for 9% and 4% of the AEM product revenue and mass, respectively. The reduced C_2H_4 fraction in the AEM configuration was offset primarily by an increased syngas fraction. O_2 is typically not considered as a product, but in this work, O_2 is the primary source of product revenue (66−67%) in both configurations. If O_2 is not sold as a coproduct, there are several CO_2RR deployment scenarios where the O_2 can be utilized in neighboring chemical processes including:[56,57] C_2H_4 oxidation to ethylene oxide,[58] vinyl acetate manufacturing (where C_2H_4, $C_2H_4O_2$, and O_2 are reacted to form vinyl acetate),[59] O_2 enrichment of iron in a blast furnace,[60] gasification,[61] and oxy-combustion of a cement clinker.[62] Rather than selling them, it may be advantageous to combust some coproducts to recover some electrical energy and offset the energy inputs for the electrolyzer.[63,64]

Comparing the cost breakdown, the AEM costs were higher for every stage except for the C_2H_5OH purification unit (Figure 2d). A 75% share of the increased costs of the AEM configuration relative to the BPM configuration originated from increased electrolyzer costs, stemming from lower area-specific C_2H_4 production rates, and the inclusion of an anodic CO_2 removal unit needed to address CO_2 crossover. These findings are consistent with other literature reports which have demonstrated that CO_2 crossover in the AEM configuration can drastically harm profitability.[20,22,65,66]

Scale Dependency of Process Cost. The BPM configuration was scaled between an industrial implementation (50,400 tonne C_2H_4/year) and lab demonstration (0.661 tonne C_2H_4/year) with financial evaluations at several intermediate sizes. As the scale was increased, the NPV became positive at 307 tonne C_2H_4/year (558 USD/tonne C_2H_4) and then peaked at 3,150 tonne C_2H_4/year (771 USD/tonne C_2H_4, Figure 3a). Further increases in scale did not enhance profitability—the 50,400 tonne C_2H_4/year had a similar but slightly reduced NPV (754 USD/tonne C_2H_4) compared to the 3,150 tonne C_2H_4/year scenario.

Scaling from a low output of 0.66 tonne C_2H_4/year to 3,150 tonne C_2H_4/year reduced the TAC by a factor of ∼450: from 1.80 million USD/tonne C_2H_4 to 3,920 USD/tonne C_2H_4 when the production costs were allocated solely to C_2H_4 (Figure 3b). At this 3,150 tonne C_2H_4/year minimum scale, the market- and mass-allocated TAC were 1,006 and 578 USD/tonne C_2H_4, respectively. As the scale was increased further to 50,400 tonne C_2H_4/year, the TAC remained relatively constant.

We divided the process costs among the different individual stages in the overall process (Figure 3c). Below 3,150 tonne C_2H_4/year, the TAC of all components (except for the CO_2 electrolyzer and cathodic CO_2 recycle heat exchangers) were the largest contributors to the TAC, and they were similar in magnitude. As the scale was increased, these process costs fell significantly and reached a plateau starting at 3,150 tonne C_2H_4/year. The electrolyzer costs remained constant at all scales. Consequently, at 3,150 tonne C_2H_4/year and 50,400 tonne C_2H_4/year, the CO_2 electrolyzer was the primary

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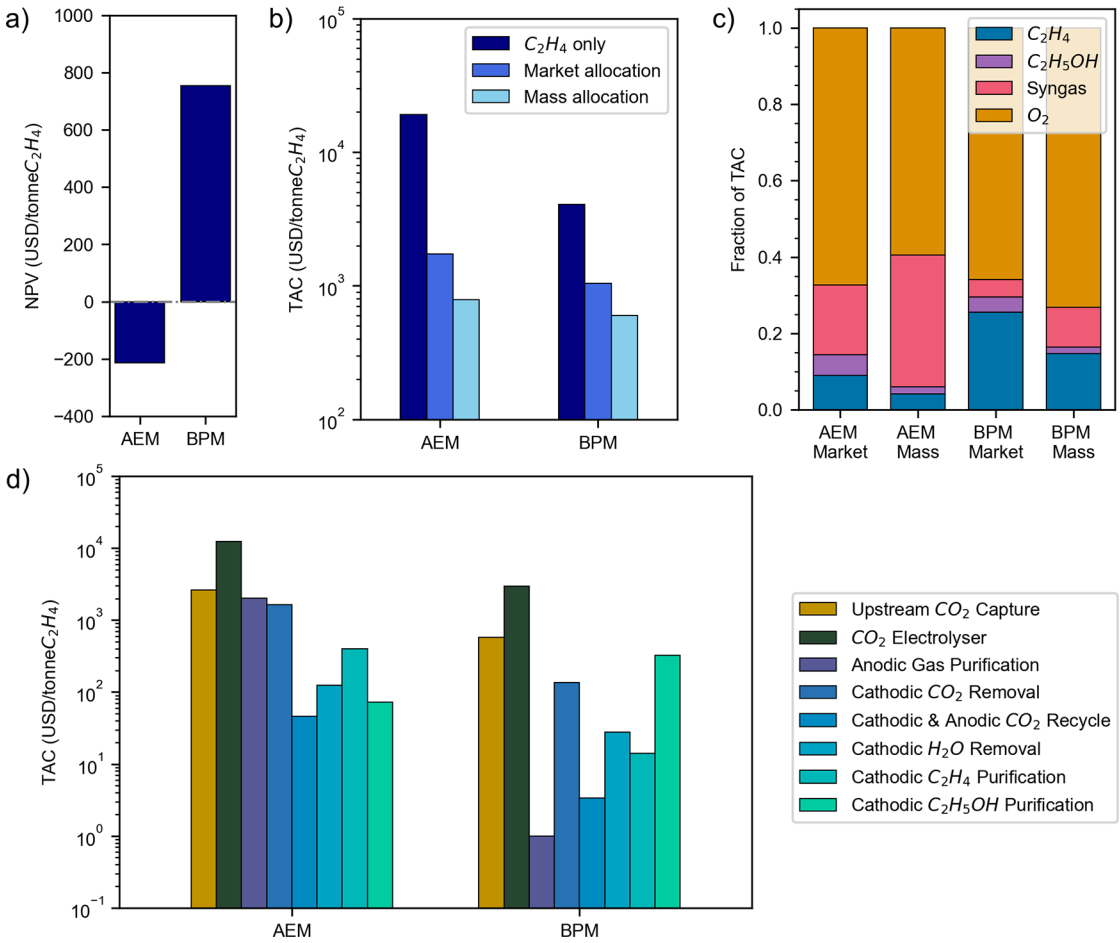

Figure 2. Comparison between the AEM and BPM electrolyzer configurations at 50,000 tonne C_2H_4/year. (a) NPV. (b) TAC with different allocation methods. (c) Fraction of TAC assigned to various products with different allocation methods. (d) Breakdown of TAC by the process stage.

contributor to the TAC (64% of the TAC at 3,150 tonne C_2H_4/year and 73% of the TAC at 50,400 tonne C_2H_4/year).

These scale optima are intrinsic to the studied process, but similar scale-up curves and economies of scale are found in other chemical processes.[67,68] The chemical processes for CO_2 removal, H_2O removal, C_2H_4 purification, and C_2H_5OH purification are based on heat exchangers and columns. The dimensions of these pieces of equipment are adjusted to achieve a desired heat or mass transfer rate, and therefore their costs scale in a nonlinear fashion. As a result, a unit designed for an order of magnitude increase in flow rate may have a slightly larger cost, but the cost may decrease when expressed on a normalized basis (i.e., per tonne C_2H_4).

The scale optimum for the integrated CO_2RR process depends on the behavior of the upstream, downstream, and electrolysis technologies, each with their own scale-up curves. The scale optimum for the overall process can change with future technological improvements (e.g., implementing an adsorbent-based approach for downstream C_2H_4 purification[69,70] or improving electrolyzer performance) or refinements to the TEA (e.g., accounting for scale-dependent electrolyzers OPEX and CAPEX[19,45,51]).

Traditional naphtha or shale gas cracking processes have plant sizes of 500,000−1,000,000 tonne C_2H_4/year.[1,2,71] Our analysis suggests that scales of only ∼3,000 tonne C_2H_4/year are needed for an integrated CO_2RR process to achieve economies of scale among the upstream and downstream process equipment. To the best of our knowledge, this is the first work to analyze the scale dependency of the costs of an integrated CO_2RR process. A 3,000 tonne C_2H_4/year facility is ∼200-fold smaller than the typical scale for the incumbent industry. The findings indicate that compared to the current fossil-fuel-based production system electrochemical C_2H_4 production has the potential to be less centralized with more distributed, modular plants.

A distributed production scheme would present challenges and opportunities for the C_2H_4 industry. The existing petrochemical supply chain is based on large-scale C_2H_4 production, and so downstream plants that act on C_2H_4 feedstocks (e.g., polyethylene plants) are often sized to match. For example, a single polyethylene manufacturing plant can consume up to 500,000 tonne C_2H_4/year,[71] although there are smaller plants (i.e., ∼70,000 tonne C_2H_4/year) in operation.[72]

Smaller CO_2RR plants are better matched to the scale of localized renewable energy installations. At 3,150 tonne C_2H_4/year, the electrolyzer would require ∼25 MW of electricity. To put this value in context, one-fifth (21%) of all utility-scale solar projects (i.e., >5 MW) in the United States have a capacity between 15 and 25 MW.[73] Alternatively, taking the average values from offshore European wind turbines in 2018 (6.4 MW power with 43% capacity factor), the electrolyzer

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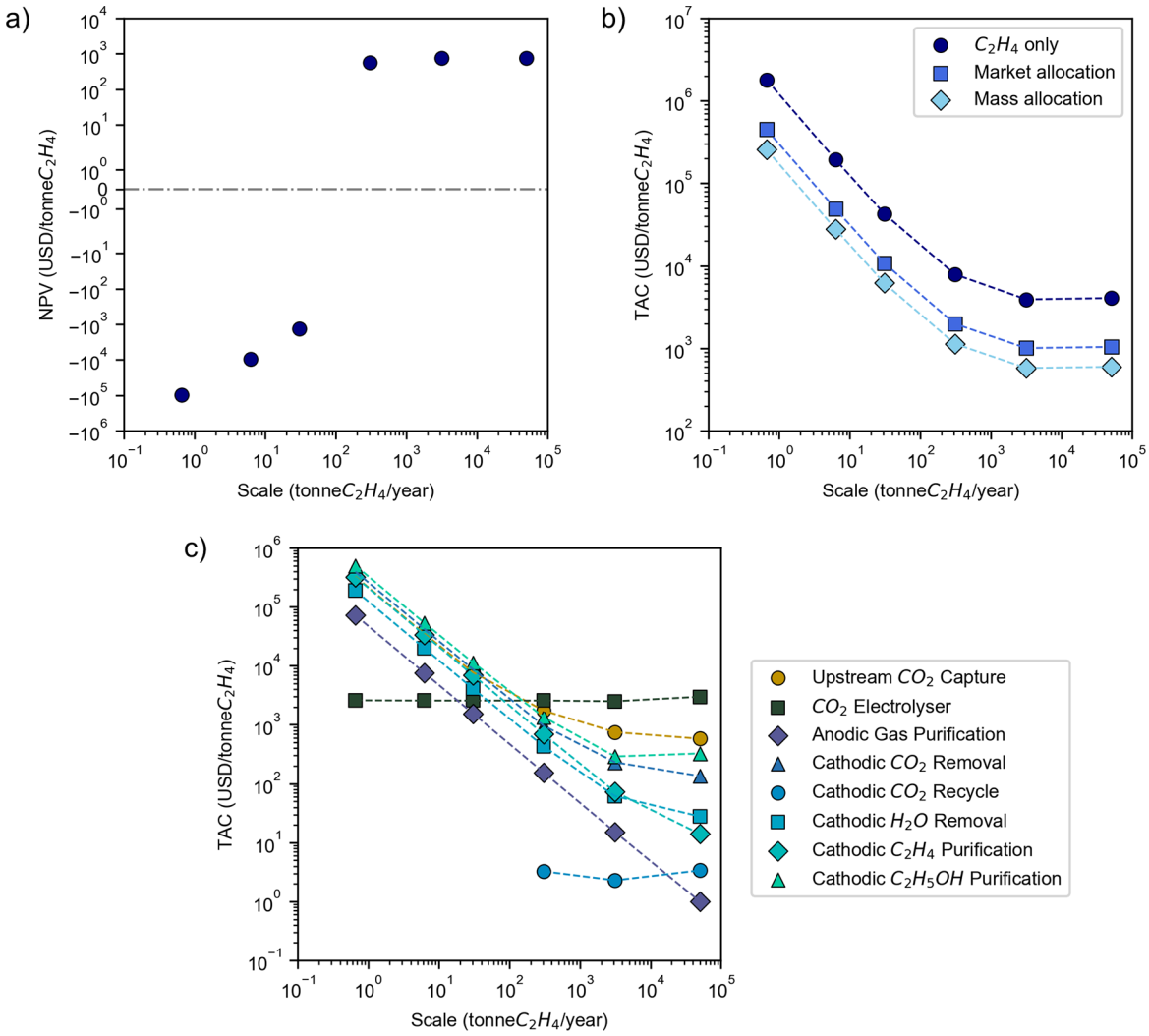

Figure 3. Effect of scale on the BPM configuration economics. (a) NPV (break-even indicated by a gray line). (b) TAC with different allocation methods. (c) Breakdown of the TAC by process stage.

could be powered by ∼9 offshore wind turbines.[74] Smaller CO_2RR plants may also be better suited to the scale of "stranded" renewable energy resources which are difficult to connect to the power grid if a proximate carbon source can be found.[75−77] Similarly, smaller CO_2RR plants would have access to more CO_2 sources.[40,78] For example, a 3,150 tonne C_2H_4/year plant would require 16,300 tonne CO_2/year, and 89% of large industrial emitters in the United States (not including power plants) could supply sufficient CO_2 for this scale.[79] A more distributed production method could also minimize the likelihood of supply chain issues across international borders, while smaller plants may reduce financial risk and facilitate easier and faster project financing, especially for the first deployment.[80]

Comparison against the Incumbent Petrochemical Process. We compared the CO_2RR to C_2H_4 process with the incumbent petrochemical process of shale gas cracking at a common scale (Supporting Information Note 5). This evaluation allows for the direct comparison of petrochemical and electrochemical processes with the same modeling assumptions (Figure S13). If the incumbent process was retrofitted with a CO_2 capture unit to recover the excess CO_2 present in the well, then the TAC exceeded that of the electrochemical conversion of CO_2 to C_2H_4 with all allocation methods. We deduce that in a low-carbon future the CO_2RR is a promising route to produce C_2H_4.

Comparison with Existing CO_2RR TEA Literature. We performed a qualitative and quantitative gap analysis to assess differences between the conclusions here and those in relevant CO_2RR literature[20−22] and to highlight reasons for discrepancies (Supporting Information Note 6). One of the primary differences that originated from our factorial method was the conservative choice of Lang factor, which substantially increased the capital costs of our system, especially for the capital-intensive CO_2 electrolyzer and the upstream CO_2 capture stages. For our downstream purification of cathodic gas streams, we implemented multiple proven technologies instead of lower energy intensity technologies that have not yet been demonstrated experimentally on CO_2RR streams.

Sensitivity Analysis. We performed a sensitivity analysis on the NPV with respect to consumables (e.g., MEA, TEG, and DI water), utility costs (e.g., electricity and steam), (co)-product market pricing, and other economic parameters (e.g., Lang factor, interest rate) at the optimal scale (Figure 4, Table S28). The CO_2 supply was varied between −30 (i.e., CO_2 transport costs via ship or pipeline)[81] and 30 USD/tonne CO_2 (i.e., a carbon price or ETS).[82] The other costs and factors were typically increased or decreased by a factor of 10 to

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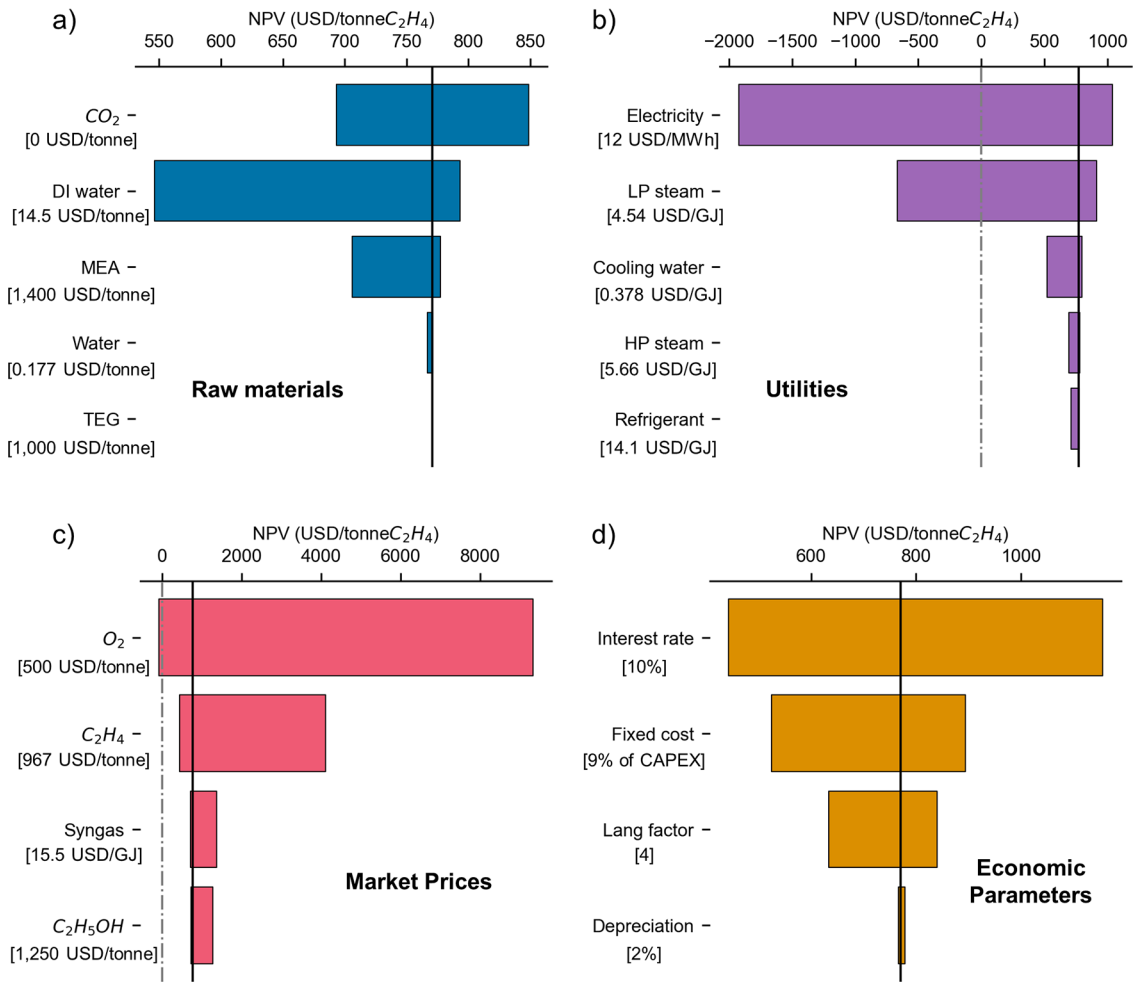

Figure 4. Sensitivity of NPV to economic assumptions (BPM configuration, 3,150 tonne C_2H_4/year). Baseline value indicated in brackets and low and high values provided in Table S28. (a) Raw material costs. (b) Utility costs. (c) Product and coproduct pricing. (d) Economic parameters.

consider uncertainties linked to the energy, chemical supply chain, and market.

Among the raw materials, the greatest reduction in NPV (−225 USD/tonneC_2H_4) occurred with a high DI water cost (Figure 4a). DI water was used extensively for electrolyte makeup, humidification, and consumption during the CO_2RR. Lower purity water was used to dilute the MEA (30 wt %). The most significant increase in NPV (+77.7 USD/tonne C_2H_4) occurred when the CO_2 supply costs became positive. However, variations in the raw material pricings, even the most prevalent ones, had a relatively minor impact on the NPV and did not change it by more than an order of magnitude.

Since we used relatively inexpensive electrolyzer electricity pricing as the baseline, a 10-fold reduction in electricity price yielded only a minor improvement in NPV (+269 USD/tonne C_2H_4), whereas a 10-fold increase drastically reduced the NPV (−2690 USD/tonne C_2H_4, Figure 4b). Since low pressure (LP) steam and, to a lesser degree, cooling water were used extensively in the downstream CO_2 recovery amine units, increases in their utility pricing considerably decreased process profitability (−1440 and −248 USD/tonne C_2H_4, respectively). Cooling water costs originated from pumping and handling the cooling water in a closed cycle—it was not consumed in the process. The utility pricing is an important consideration as increased electricity and LP steam costs can result in a loss of process profitability (i.e., a negative NPV).

Improving the (co)product market price improved the profitability of the overall process (Figure 4c). If the anodic product, O_2 in this case, increased 10-fold in economic value, the NPV improved substantially (+8560 USD/tonne C_2H_4). A decreased O_2 market price was the only market price which could jeopardize the process profitability (−856 USD/tonne C_2H_4). Increasing the market price of C_2H_4 made the process significantly more profitable (+3330 USD/tonne C_2H_4), whereas increases in C_2H_5OH and syngas prices only slightly increased the NPV (+507 USD/tonne C_2H_4 and +592 USD/tonne C_2H_4, respectively).

The economic parameters are subjective, and hence, they were included in the sensitivity analysis (Figure 4d). The interest rate was the most impactful economic parameter varied (+385 USD/tonne C_2H_4, −328 USD/tonne C_2H_4). The second most significant parameter was the fixed operating costs (i.e., maintenance, taxes and insurance, factory and head office overhead, and labor) which increased or decreased the NPV by +123 or −246 USD/tonne C_2H_4, respectively. The Lang factor (ratio between the facility CAPEX and the cost of the major process equipment) and the salvage value declining depreciation rate (capital value of equipment loss during operation) did not have a significant impact on the NPV at this scale.

Roadmap to Improved CO_2RR Profitability. Cumulative improvements in the techno-economic parameters can

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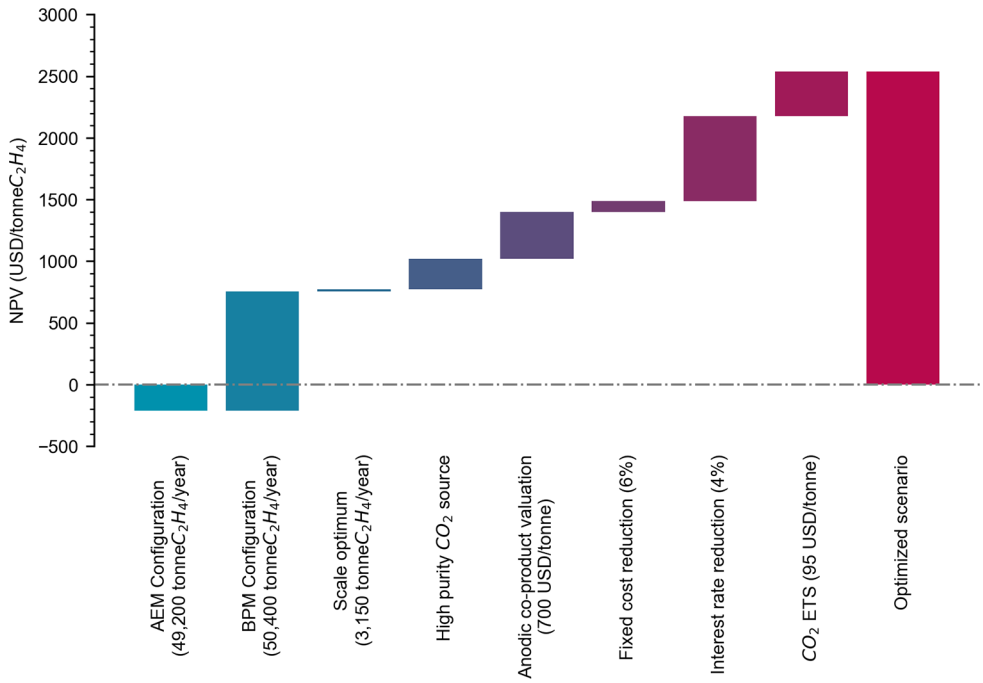

Figure 5. Roadmap to improve CO_2RR profitability via improvements in technical, deployment, and economic factors (break even indicated by the gray line). The mass allocated TAC with the same improvements is plotted in Figure S15.

greatly improve the profitability of the CO_2RR process. We explored the financial implications if several advancements were realized sequentially in one industrial deployment (NPV plotted in Figure 5, mass allocated TAC in Figure S15). Transitioning to the BPM electrolyzer configuration and reaching the optimum scale were critical to enable the CO_2RR process to break even (i.e., reach a positive NPV at 771 USD/tonne C_2H_4). To be conservative, we selected a very dilute CO_2 source (6.4 wt % or 4.2 mol %), but the capture cost is proportional to the CO_2 content in the flue gas.[40] Other works have demonstrated that increasing the CO_2 concentration from ∼4 mol % to ∼24 mol % can substantially reduce CO_2 capture costs: from 75 to 100 USD/tonne CO_2 to 31−35 USD/tonne CO_2.[40] If a high-purity CO_2 source is available (e.g., C_2H_5OH fermentation provides CO_2 at ≳98 wt %[40,83]), then the upstream CO_2 capture stage could be completely eliminated, and the NPV would increase by 247 USD/tonne C_2H_4. Adapting the anodic reaction to produce a more valuable chemical commodity than O_2 (i.e., 700 USD/tonne), without harming the cathode performance, greatly benefited the NPV (+380 USD/tonne C_2H_4). Therefore, alternative anodic reactions may be worth exploring if inexpensive feedstocks can be secured.[84,85] The NPV was further improved by reducing the fixed operating costs (from 9% to 6% of the CAPEX, +90 USD/tonne C_2H_4). Practically, this could be realized by collecting incentives and insurance bonuses, by incorporating automation and designing the CO_2RR process to minimize maintenance, and by gaining access to low-cost land.[86] Access to financial institutions with lower interest rates (4%) and higher CO_2 value (95 USD/tonne CO_2—the EU average ETS in 2022[87]) were also important factors (+689 USD/tonne C_2H_4 and +360 USD/tonne C_2H_4, respectively). Following this roadmap, the NPV reached as high as 2,540 USD/tonne C_2H_4, more than triple the NPV at the scale optimum for the BPM configuration.

CONCLUSION

We carried out a detailed scale-dependent TEA (−30%, +50% accuracy) of CO_2 capture, CO_2RR, and downstream purification of C_2H_4, syngas, O_2, and C_2H_5OH. With a BPM electrolyzer configuration operating at the optimum scale (3,150 tonne C_2H_4/year) the TAC was estimated at 3,920 USD/tonne C_2H_4 (C_2H_4 only) or 578 USD/tonne C_2H_4 (mass allocation), with an NPV equal to 771 USD/tonne C_2H_4. A production rate of ∼3,000 tonne C_2H_4/year was sufficient to achieve economies of scale in the upstream and downstream chemical processes, suggesting that the CO_2RR to C_2H_4 can be used for distributed production schemes. Future work can explore how the costs and performance of CO_2 electrolysis on different scales impact these findings. Intermittent renewables and imperfect capacity factors should also be considered to further develop a realistic industrial model. We determined some short-term technical and economic enablers that improved the CO_2RR process economics and would make it more profitable: accessing cheap high-purity CO_2 feedstocks, reducing fixed operating costs, accessing lower interest rates, and implementing CO_2 pricing. With these enablers, the NPV climbed as high as 2,540 USD/tonne C_2H_4. We conclude that the electrochemical conversion of CO_2 to C_2H_4 could provide an economically viable avenue for C_2H_4 production and a viable alternative to the incumbent petrochemical processes. This analysis, complete with scale-dependent upstream and downstream analysis, provides guidance for the scale-up of this technology.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.3c04373.

Selection of separation technology; Aspen Plus model specifications; CO_2 electrolyzer performance; C_2H_4 adsorption in the downstream CO_2 recovery unit with

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different capture solutions; CO adsorption in the downstream CO_2 recovery unit at different solid oxide electrolyzer cell (SOEC) production rates; ratio of purified C_2H_4 to electrochemically produced C_2H_4; techno-economic model; Aspen Plus model of the benchmark process; TAC of C_2H_4 production from CO_2RR compared to the SG benchmark process with and without CO_2 capture; comparison to CO_2RR TEA literature; values used in the sensitivity analysis; roadmap to improve CO_2RR profitability via improvements in technical, deployment, and economic factors (PDF)

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●T.A., A.G., and J.P.E. contributed equally to this work.

Notes

The authors declare the following competing financial interest(s): C.M.G., C.P.O., and D.S. have a financial interest in CERT Systems Inc., a company commercializing CO_2 electrolysis technology. C.M.G., C.P.O., and D.Y. are employees of CERT Systems Inc. J.W. is an employee of Twelve Benefit Corporation, a company commercializing CO_2 electrolysis technology.

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ABBREVIATIONS

C_2H_4, Ethylene; CO_2, Carbon dioxide; C_2H_5OH, Ethanol; CO_2RR, Carbon dioxide electroreduction; TEA, Techno-economic assessment; TRL, Technology readiness level; CO, Carbon monoxide; H_2, Hydrogen; O_2, Oxygen; MEA, Monoethanolamine; N_2, Nitrogen; H_2O, Water; AEM, Anion exchange membrane; BPM, Bipolar membrane; $KHCO_3$, Potassium bicarbonate; DI, Deionized; CH_2O_2, Formic acid; $C_2H_4O_2$, Acetic acid; C_3H_7OH, 1-Propanol; TEG, Triethylene glycol; CAPEX, Capital expenditures; OPEX, Operating expenditures; NPV, Net present value; TAC, Total annualized cost; ETS, Emission trading system

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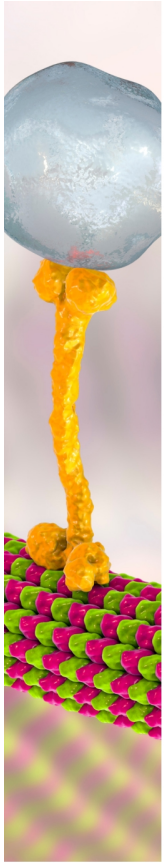
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